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A state-of-the-art survey of deep learning models for automated pavement crack segmentation

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ABSTRACT

Survey of road cracks in a timely, complete, and accurate way is pivotal to pavement maintenance planning. Motivated by the increasingly heavy task of identifying cracks, researchers have developed extensive crack segmentation models based on Deep learning (DL) methods with significantly different levels of accuracy, efficiency, and generalizing capacity. Although many of the models provide satisfying detection performance, why these models work still needs to be determined. The objective of this study is to survey recent advances in automated DL crack recognition and provide evidence for their underlying working mechanism. We first reviewed 54 DL crack recognition methods to summarize critical factors in these models. Then, we conducted a performance evaluation of fourteen famous semantic segmentation models using the quantitative metrics: F-1 score and mIoU. Then, the effective receptive field and class activation map of the included models are visualized to demonstrate the training results as qualitative evaluation. Based on the literature review and comparison results, larger kernel size, feature fusion, and attention module all contribute to the improvement of model performance. Striking a balance between increasing the effective receptive field and computational/memory efficiency is the key to designing DL crack segmentation models. Finally, some potential directions and suggestions for future development are provided, such as developing semi-supervised or unsupervised learning for the high cost of pixel-level labeling.

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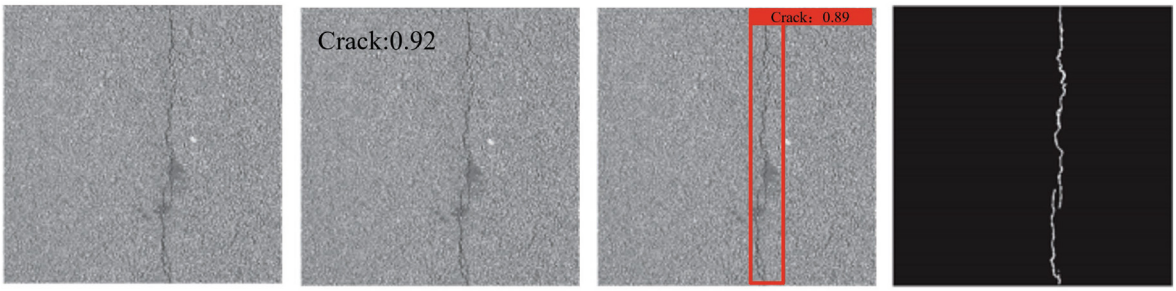
1. Introduction

Cracks decrease the durability of the pavement and accelerate the deterioration process. Therefore, a timely, complete, and accurate survey of road pavement conditions is pivotal to pavement maintenance planning (Adlinge and Gupta, 2013). With the development of image acquisition approaches, automated road condition monitoring technology has been widely applied, resulting in a massive dataset of disease images. Among all types of pavement surface distress data, crack images are the hardest to process with satisfying accuracy and efficiency. The increasingly heavy task of identifying cracks motivates researchers to propose crack detection methods.

Automated crack detection methods consist of two procedures: image acquisition and crack recognition. Data acquisition provides two- or three-dimensional crack image data using cameras, smartphones, or other tools. Crack recognition requires distinguishing pixels or patches containing cracks from the complex background and generating category labels, bounding boxes with positioning information, or pixel-level masks. Approaches to automating crack recognition can be divided into three categories: crack recognition based on image processing techniques, machine learning methods with artificial design features, and ones based on deep learning.

Based on image processing methods, various technologies can be applied to identify, locate and classify pavement diseases, such as using edge information (Abdel-Qader et al., 2003), morphological operations, thresholding (Cheng et al., 2003), digital image correlation (Gehri et al., 2020), pattern matching (Adhikari et al., 2014) and filtering algorithms (Zhang et al., 2013) to identify cracks. Traditional image processing methods offer advantages in specific scenarios, such as simplicity, efficiency, and interpretability. However, they may struggle to handle complex scenes and unstructured data. Machine learning-based (ML) methods have also been widely applied to detect cracks. By recognizing artificial features designed in advance by experts, such as geometric features, texture, color, and grayscale information of the crack in the image, a machine learning-based algorithm could determine whether an image contains a crack (Oliveira and Correia, 2012; Amhaz et al., 2016a; Amhaz et al., 2016b; Shi et al., 2016; Gong et al., 2019; Chen et al., 2017a). Although ML methods have achieved great detection results, their performance depends heavily on artificially designed features and lacks universality, making it difficult to achieve real automatic processes.

In recent years, deep learning (DL) algorithms have made tremendous breakthroughs. Compared to traditional image process methods and ML algorithms, DL algorithms can automatically learn meaningful feature representations from images and handle complex scenes and large-scale data. Inspired by their remarkable successes in image processing, DL-based approaches have also been applied to tackle pavement crack recognition tasks that can be divided into three task categories: image classification, object detection and semantic segmentation as shown in Fig. 1. The image classification approaches require the model to determine whether an image or patch contains a crack (Gopalakrishnan et al., 2017; Zhang et al., 2016; Zhang et al., 2022b). In contrast, the object detection approaches use one-stage methods (Mandal et al., 2018; Ukhwah et al., 2019; Du et al., 2021) or two-stage methods (Nie and Wang, 2018) to localize targets and generate bounding



(a) Origin Image (b) Image Classification (c) Object Detection (d) Semantic Segmentation

Fig. 1. Different DL crack recognition approaches.

boxes. The semantic segmentation approaches further segment pixels of different targets from background pixels based on convolutional neural networks (Panella et al., 2018) or Transformer (Liu et al., 2021a).(see Fig. 2)

By investigating different types of crack recognition approaches, methods based on deep learning outperform others in terms of accuracy and automation. Pavement condition ranking requires both the impacted area and the severity of cracking, which makes the accuracy of both the detected crack pattern and width essential. Only DL semantic segmentation can achieve pixel-level recognition accuracy and output geometry information of cracks that can be utilized to evaluate the state of pavement deterioration more practically. Therefore, crack segmentation has become the mainstream in the field of crack recognition.

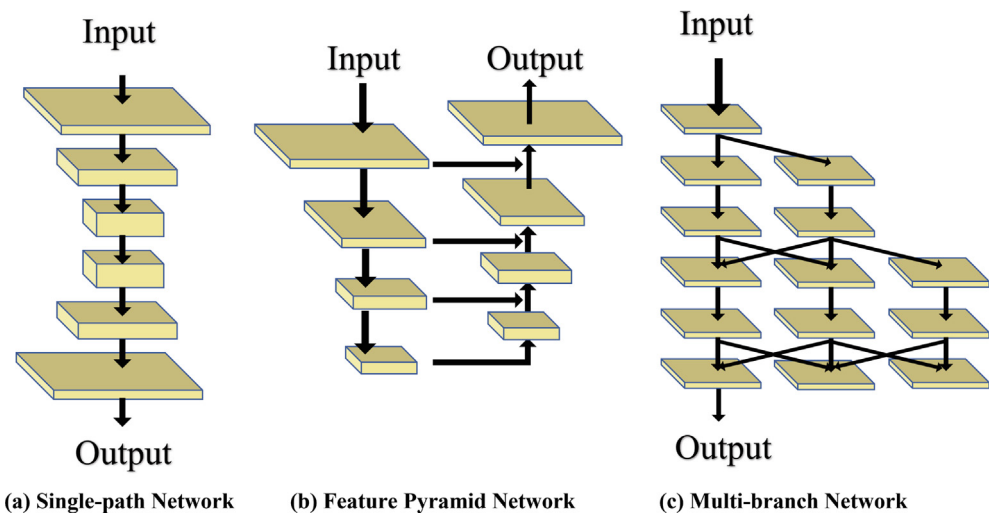
2. Overview of existing models

The previous review of crack segmentation focused on listing the different types of approaches and comparing their differences, failing to explain why these models work from a model design perspective. This section presents a comprehensive overview of existing DL-based crack segmentation models and summarizes key components contributing to their performance to provide a little guidance for the construction of the crack segmentation model.

The Encoder-Decoder structure can be considered as a paradigm for crack segmentation models. In these architectures, encoders such as VGG, ResNet, Transformer, and their variants are commonly used for feature extraction, while the decoder progressively recovers object detailed information and spatial dimensionality. According to the types of encoder and decoder, DL-based models can be divided into two main types: Models based on convolutional neural networks or attention mechanisms.

2.1. Model based on convolutional neural networks

As a specialized type of artificial neural network, CNNs use convolution instead of general matrix multiplication in at least one of their layers (LeCun et al., 2015). They are specifically designed to process image data, widely used in image recognition



(a) Single-path Network **(b) Feature Pyramid Network** **(c) Multi-branch Network**

Fig. 2. Illustration of different CNN structures for image segmentation.

and processing due to their inductive bias: weight sharing and local connectivity. According to the shape of these network structures, these models can be divided into single-path network, feature pyramid networks and multi-branch parallel network.

2.1.1. Single path network

A single-path network contains only one mainstay in the structure. Just like FCN (Long et al., 2015), the input image first passes through multiple downsampling layers, thus gradually expanding the receptive field to obtain higher-level features containing more abstract semantic information, then transforming the high-dimensional feature maps into multiple probability maps of the same size as the original size by bilinear interpolation. The probability maps indicate the probability value that the pixel belongs to a particular category, thus achieving pixel-level image segmentation. To refine the segmentation boundary, skip connection (Ronneberger et al., 2015) and position index (Badrinarayanan et al., 2017) are added to preserve the spatial information before downsampling, which inspired us that lateral output and feature fusion can be used to reuse the encoder's high-resolution feature mapping to recover the lost spatial information.

2.1.2. Feature pyramids network

In light of Feature Pyramid Network (Lin et al., 2017), the focus of semantic segmentation began to shift from increasing network depth to exploiting image context information and extracting multi-scale features more effectively. To build feature pyramids by resizing and concatenating specific feature maps of different layers, thus improving the utilization of shallow feature maps through feature fusion and providing multi-scale feature maps with receptive fields in different scales. Feature pyramids allow the model to capture both local and global context information, which is helpful for accurately recognizing cracks of different sizes and shapes. Chen et al. (2017b) designs an Atrous Spatial Pyramid Pooling module (ASPP) to capture the multi-scale contextual information in Deeplabv2, while Zhao et al. (2017a) employs a Pyramid Pooling module (PPM) in PSPNet to fuse the contextual information. Based on the PPM, Xiao et al. (2018) further builds feature pyramids to capture more contextual information at various levels in UperNet.

2.1.3. Multi branch network

Most CNN-based crack segmentation models may suffer from loss of spatial information in the high-to-low and low-to-high resolution processes. Therefore, some researchers propose multi-branch parallel models for crack segmentation, which separates spatial information and semantic information by multi-branch convolutional flow to achieve a balance between classification accuracy and localization precision. For example, Wang et al. (2020) maintains high-resolution representation throughout the whole process through parallel multi-resolution convolutions. BiseNet encodes spatial information and semantic information through two paths, respectively. These models achieved state-of-the-art performance owing to their novel structure design (Yu et al., 2018).

Methods		Model Modification
Single path	FCS-Net	Li et al. (2021) applied a pre-trained ResNet-50 with batch normalization and atrous spatial pyramid pooling
	CrackSegNet	Ren et al. (2020) designed a modified VGG-16 architecture with dilation convolution and spatial pyramid pooling.
	U-HDN	Song et al. (2020) proposed an encoder-decoder architecture with a multi-dilatation module and a hierarchical feature learning module.
	CrackW-Net	Based on U-Net, a skip-level round-trip sampling block is added in CrackW-Net (Han et al., 2022).
Pyramid	DeepCrack [Liu]	The side-output layers with deep supervision are concatenated to form a final fused output of DeepCrack (Liu et al., 2019).
	DeepCrack [Zou]	SegNet's variants with feature fusion in each convolution stage (Zou et al., 2019).
	FPHBN	Yang et al. (2020) introduced a feature pyramid module and a hierarchical boosting module integrated into HED (Xie and Tu, 2015).
	SDDNet	SDDNet (Choi and Cha, 2019) consisted of a DenSep modules, a modified ASPP module, and a decoder module.
Multi branch	HRNet[e]	HRNet's variants with the addition of dilated convolution and hierarchical feature integration (Chen et al., 2021).
	EHRS-Net	Xu et al. (2022) designed three different semantic representation extraction flows exchanging semantic information in parallel.
	BiseNet	BiseNet (Wang and Su, 2021) consists of two paths: spatial path and context path, that are fused after the attention refinement module.
	ShuttleNet	Zhang et al. (2022a) stacks encoding-decoding round to increase reuse of feature maps.

2.2. Fundamental components in CNNs models for crack segmentation

According to the analysis of existing CNN-based models, some fundamental components in CNNs models for crack segmentation are discussed as follows: model structure, convolution block design, activation function, and other details like normalization methods.

2.2.1. Network Structure

CNNs structure the direction and pattern of semantic information flowing and interacting, with both network depth and shape having a critical impact on model performance.

Network Depth. Stacking multiple convolutional layers with a small filter to form a network model such as VGG (Simonyan and Zisserman, 2014) and ResNet (He et al., 2016) can effectively expand the receptive field. Deeper networks always mean more robust representation learning capabilities, as well as more computational load. In crack segmentation models, however, the depth is always limited for localization accuracy (downsampling multiplier at most five times).

Network Shape. As discussed above, the shape of the network can be divided into three types. Multi-branch parallel networks and feature pyramid networks always achieve higher performance, while a complex structure inevitably leads to excessive memory consumption. Single-path networks are inferior in segmentation accuracy but easier to train and deploy. Therefore, it is necessary to choose the appropriate model structure according to the actual engineering needs.

2.2.2. Convolution block design

Convolutional neural networks are usually stacked with multiple convolution blocks, which are groups or sequences of layers designed to extract features from input data using convolution operations. A convolution kernel, also known as a filter, represents its encoding from some aspect of the input data. The shape and number of kernels in each layer significantly affect the feature map.

Kernel size. The size of the convolutional kernel is closely related to the receptive field of the neuron. Recently, with the development of the hardware, more researchers have gradually applied larger kernels to improve the segmentation performance instead of stacking 3×3 small kernel. RepLKNet (Ding et al., 2022), a pure CNN architecture using 31×31 kernel size, significantly closes the performance gap between CNNs and ViTs. Liu et al. (2022) applied the same concept when designing the ConvNet, further demonstrating the tremendous potential of large kernel convolution in the field of crack segmentation.

Filter number. Most abstract features extracted from the high-dimensional feature maps are unintelligible in the images, so more convolutional kernels are necessary to resolve and distribute these image replicas into different channels. The number of kernels directly determines the capacity and depends on the number of available examples and the complexity of the task. Currently, there is yet to be a universally accepted standard for the number of filters, and many researchers rely on continuous parameter tuning and refinement and their own experience to determine the appropriate number of filters.

Convolution type. In addition to the size and number of convolution kernels, the type of convolution has a significant impact on the performance of the model. Besides the standard convolution, there are group convolution, dilated convolution (atrous convolution), and separable convolution. By grouping the convolution operations, it reduces the computational cost and the number of parameters. Dilated convolution expands the receptive field by introducing gaps within the kernel. In contrast, depthwise separable convolution is commonly used in lightweight models and mobile devices to improve computational efficiency and reduce model size.

2.2.3. Activation function

The role of the activation layer is to introduce nonlinearities in the overall network without affecting the receptive fields of the convolutional layers. Taking the linear arrays obtained by passing the neurons in the previous layer and processing them through various nonlinear activation functions to achieve the simulation of arbitrary functions, thus completing the purpose of enhancing the representative capability of the network.

There are different usage scenarios for other activation functions. Sigmoid and ReLU activation functions are widely used in crack segmentation models due to their ease of computation and faster convergence. In recent years, more practical activation functions such as Leaky ReLU (Maas et al., 2013), ELU (Clevert et al., 2015) and GeLU (Devlin et al., 2018) are proposed to increase the nonlinearities in these models. GELU precisely introduces the idea of random regularity in activation, thus achieving top performance.

2.3. Models based on attention mechanism

The idea behind the attention mechanism is to allow the decoder to flexibly exploit the most relevant parts of the input sequence by a weighted combination of all encoded input vectors, with the most pertinent vectors receiving the highest weights. The attention mechanism can be seen as a special distribution of weights.

The attention mechanism mainly used in segmentation can be divided into self-attention, channel attention, and spatial attention. Self-attention is usually used for feature extraction as the encoder of segmentation models, while spatial and channel attention are integrated as modules to modify the visual field distribution. According to the components in the network, models based on the attention mechanism can be divided into pure Transformer-based models and attention-integrated CNNs models.

2.3.1. Pure Transformer-based models

A transformer is a DL model that utilizes the self-attention mechanism to capture relationships between words or elements in a sequence, enabling practical language processing and generation tasks (including the recursive output). The self-attention mechanism generates the corresponding Query, Key, and Value matrix based on the input data. Owing to its outstanding capability to capture internal correlations in image data, researchers gradually propose a variety of image-processing methods based on the self-attention mechanism. Wang et al. (2018) proposed the Non-Local module to build the contextual relationship through the self-attention mechanism to effectively capture long-range feature dependencies. In 2020, the Vision Transformer ViT) (Dosovitskiy et al., 2020) was proposed and outperformed the CNN through transfer learning. Swin-Transformer (Liu et al., 2021b) improved computational efficiency by using small sub-networks and structured skip-connections, allowing better utilization of information across layers. Strudel et al. (2021) formally introduced ViTs to the field of semantic segmentation in Segmenter, using Transformer instead of convolution as the backbone to capture contextual information, and showed novel segmentation performance. Transformer, with its excellent performance, has also won the favor of many researchers in the field of crack segmentation, promoting the development of Transformer-based crack segmentation methods (Liu et al., 2021a; Kang and Cha, 2022; Guo et al., 2023).

Purely transformer-based models have proven more powerful than convolutional neural networks in feature extraction for the classification task. However, their limitations are apparent. For example, Transformer-based methods focus more on capturing long-range dependency information but lack local perception, which easily leads to incorrect prediction of small targets. Furthermore, a large receptive field too early results in redundant memory consumption. Transformers can only achieve reliable performance by pre-training with large datasets, making training of Transformer-based models costly. In addition, there is a direct gap between classification and segmentation, so it is worth further exploring how to design a better transformer structure conducive to segmentation.

2.3.2. Attention integrated CNNs models

Both CNNs and pure Transformers have excellent performance in computer vision, and the combination of them can sometimes achieve even better results in crack segmentation. According to the literature review, the attention mechanism has become a critical module to improve the performance of CNN. There are usually three types of attention mechanisms used in CNNs: spatial attention (Jaderberg et al., 2015) and channel attention (Shen et al., 2021; Hu et al., 2018) and their combination, such as Dual Attention Network (Fu et al., 2019) and Convolutional Block Attention Module (CBAM) (Woo et al., 2018), which focus on the spatial region and channel levels, respectively or both. Unlike the self-attention mechanism, these attention modules focus on redistributing the existing weights. The combination with attention modules can effectively improve the performance of crack segmentation models (Xiang et al., 2020; Pan et al., 2020; Qu et al., 2021).

3. Critical factors to model performance

Semantic segmentation refers to a fundamental CV task that aims to recognize each pixel of the given images according to a set of predefined semantic categories. The performance of the crack segmentation models can be reflected in multiple aspects, such as recognition precision, localization accuracy, processing speed, and generalization ability. The critical factors of model performance and evaluation methodologies will be discussed in this section.

3.1. Tradeoff between semantic precision and spatial accuracy

As discussed above, it is essential to properly handle the trade-off between classification and localization. For single-pixel classification, the neural network must learn as much condensed semantic information associated with the target pixel as possible, thus focusing more on global information. The localization task, on the other hand, should pay more attention to local spatial location information to determine the relative relationship between different pixels.

In particular, segmenting cracks of different sizes requires different scales of contextual information due to the complex genesis, variable shapes of cracks, and various shooting angles. By summarizing and comparing an array of excellent backbone and crack segmentation models, the key to achieving a trade-off between semantic precision and spatial accuracy is to broaden the receptive field and feature fusion.

3.2. Receptive fields

In a neural network, each neuron receives input only from a limited area, which is called the receptive field. Fig. 3 will help to understand the RF. The representations extracted by the neural network represent the local area of the input and possess a small receptive field, while the deeper layer corresponds to a larger receptive field that can reflect more global information about the original input. As shown in Fig. 3, it can be assumed that the larger the receptive field, the stronger the ability to extract features and resist interference. The formula for calculating the theoretical perceptual field is as follows, where l_{k-1} is the corresponding receptive field size at the (k-1)th layer:(see Fig. 4)

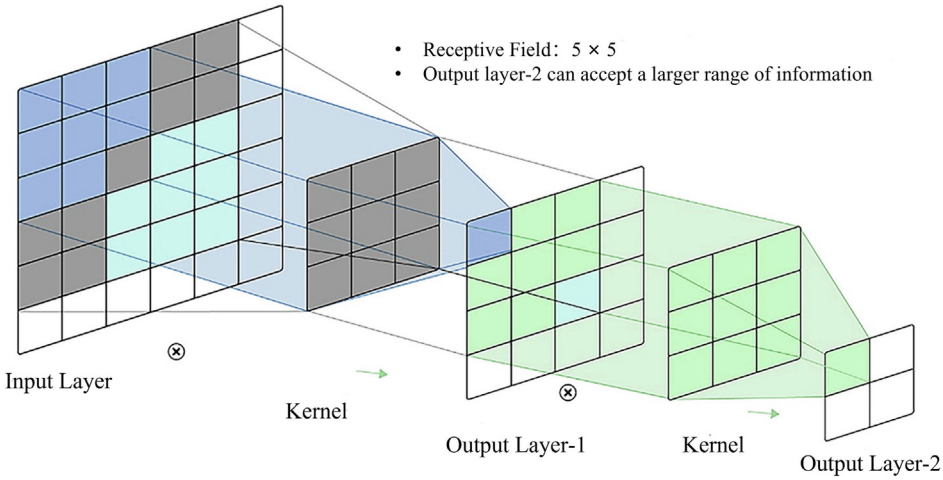


Fig. 3. Theoretical receptive field.

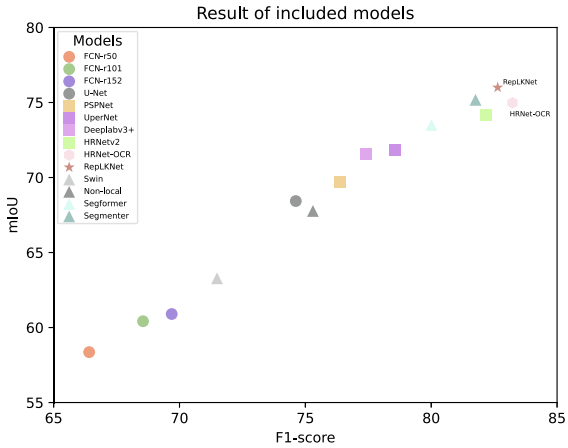


Fig. 4. Comparision result. The closer the data to the top right, the better the model's performance.

$$l_k = l_{k-1} + [(\text{Size}_{\text{kernel}} - 1) \times \prod_{i=1}^{k-1} \text{Stride}] \tag{1}$$

There is an apparent gap between the theoretical receptive field (ERF) and the actual effective receptive field. Luo et al. (2016) proposed that the actual effective receptive field obeys a Gaussian distribution and exhibits a much smaller area than the theoretical receptive field. When the network structure is determined, the corresponding property of the effective receptive field is determined. Therefore, only by increasing the ERF can the performance of the model be effectively improved.

Depending on the literature review, there are several alternatives for broadening the ERF, which can be divided into three types: 1) Established techniques; 2) Feature Fusion; 2) Attention Mechanism.

Traditional techniques. The traditional techniques for enlarging the receptive field include augmenting the size of convolutional kernels, raising the network depth, or applying the dilation convolution. The limitation of this type of method is that a large but single-scale receptive field often performs below expectations and even exhibits misleading answers when the target objects are detected with multiple scales. Thus, it is of utmost importance to obtain a multi-scale receptive field.

Feature fusion. In CNNs, feature fusion is defined as combining feature maps from different levels or branches, which helps to learn multi-scale representations and long-range dependencies. According to the scope of its implementation, feature fusion can be divided into local fusion and global fusion. Local fusion, which occurs only within a subnetwork module or bottleneck, outputs a feature map that fuses features with multiple sizes and multi-scale receptive fields. There are numerous convolution blocks based on local fusion, such as SPP Module (He et al., 2015), ASPP (Chen et al., 2017b), Receptive Field Block (Liu et al., 2018), PPM (Zhao et al., 2017a), and so on. Compared to local fusion, global fusion emphasizes the fusion of deep semantic features and shallow spatial information. Typical examples of such fusion methods are various segmentation models such as HED (Xie and Tu, 2015), FPN (Lin et al., 2017), UperNet (Xiao et al., 2018), and HRNet (Wang et al., 2020).

Attension mechanism. Compared to CNNs, the main advantage of the attention mechanism is its ability to capture global contextual information, which is equivalent to extending the receptive field to the whole image. According to the experimental results, the shallow networks of the Vision Transformer can capture a broader range of global information, which contributes to the improvement of classification accuracy. Without the need for down-sampling, ViTs preserve as much spatial information as possible, which is conducive to localization.

3.3. Image resolution and training sample size

The dataset used to train and test the crack segmentation model is critical to model performance. The sample size directly determines the accuracy and robustness of the model. One of the current challenges faced by crack segmentation primarily stems from the poor generalization of models trained on small-scale datasets. Fast, robust, and accurate data acquisition and ground-truth labeling should be further developed.

Image resolution is also an important factor that affects the computational complexity and performance of CNNs. For the same network, higher resolution typically leads to higher FLOPs and higher accuracy. Models trained on low-resolution images perform poorly at detecting thin cracks at a relatively large distance from the camera and smartphone.

3.4. Evaluation metrics

For crack segmentation, performance measurement is also an essential task. However, most existing studies use precision, recall and accuracy as metrics, failing to strike a tradeoff between false positive and false negative samples because of class imbalance. The intersection over union (IoU) and F-1 score can consider false positive and negative detections, making it widely used in crack segmentation. In addition, many studies in deep learning have also proposed more reasonable metrics to efficiently evaluate their models, including

Optimal dataset scale and optimal image scale. Typically, the crack segmentation output will be grayscale images with pixel values between [0, 255]. It is necessary to find an optimal threshold that can binarize the edge map output to distinguish between background and cracks. Based on the binary maps, we can then compare the prediction to the ground truth to compute metrics such as precision, recall, and F-score (Liu et al., 2017). There are two ways to compute this optimal threshold and F-score:

- ODS: Iterate over all possible threshold values and set it as a common threshold for the entire dataset. Whichever threshold gives you the best F-score for the dataset becomes the ODS F-score.
- OIS: For each image, calculate the best threshold and corresponding F-score. Average out each F-score for each image, and that becomes the OIS F-score.

4. Evaluation of different models on open dataset

To verify the effect of different structure designs and other critical factors on model performance, we conducted this comparative experiment. In this section, we first introduce the involved datasets, included models, and implementation details settings. Then, we report the crack segmentation results using the quantitative metrics: F1 and mIoU. The effective receptive field and class activation map of the included models are visualized to demonstrate the results as a qualitative evaluation. Finally, some analysis and discussion are presented.

4.1. Datasets

In recent years, the emergence of numerous pavement crack datasets with pixel-wise annotation has extensively promoted the progress of crack segmentation. Five public crack datasets are used in this research, including Crack500 (Yang et al., 2020), CrackTree260 (Zou et al., 2012), CrackSL315 (Zou et al., 2019), CrackWH100 (Zou et al., 2019), and Stone331 (König et al., 2021). To further verify the performance of different methods under a complex environment, we combine all the datasets together and establish the C4S dataset. This combined dataset has difficulties like shadows, occlusions and noise, etc., which challenges the generalization ability and robustness of the model. 70% of the dataset was used as the training set, 20% was used as the validation set, and 10% was used as the test set.

4.2. Implementation details

For implementation details, all networks are PyTorch-based implementations, with PyTorch-1.13.1 + CUDA-11.6. All experiments are run on two NVIDIA RTX 3090 GPUs. Each model is trained for 100 epochs. During training, the initial global learning rate is set to 1e-4. Weight decay and momentum are set to 0.0005 and 0.9, respectively. The stochastic gradient descent (SGD) method is used to update the network parameters with a batch size of eight (the largest batch size RepLKNet is compatible with) in each epoch.

The objective of this experiment is to conduct a direct comparison among various models under the same benchmarking conditions. All parameter settings, data augmentation methods, and training strategies are consistent with the default con-

figuration in MMSegmentation (a standardized semantic segmentation framework based on the PyTorch implementation) (Contributors, 2020).

4.3. Included models

We compare the performance of some classical methods. In accordance with the comparison, we can find the actual influencing factors to crack segmentation performance. CNNs-based models includes FCN (Long et al., 2015), U-Net (Ronneberger et al., 2015), UperNet (Xiao et al., 2018), PSPNet (Zhao et al., 2017b), Deeplabv3+ (Chen et al., 2017), HRNet (Wang et al., 2020), RepLKNet-31 (Ding et al., 2022) and HRNet-OCR while attention-based models includes Non-local (Wang et al., 2018), Swin (Liu et al., 2021b), Segformer (Xie et al., 2021) and Segmenter (Strudel et al., 2021).

4.4. Quantitative evaluation

For each image, Precision and Recall can be calculated by comparing the detected cracks against the human-annotated ground truth. Then, the F-measure ($\frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$) and mIoU (Mean Intersection over Union) can be computed as an overall quantitative metric for performance evaluation.

4.5. Qualitative evaluation

In addition to quantitative metrics, qualitative evaluations: the class activation map (CAM) and the effective receptive field (ERF) of the included models are visualized to further understand the training results of different models.

For a given class (or category), the CAM indicates the discriminative region of the image that influenced the deep learning model to make the decision. In this study, we apply gradient-weighted class activation mapping (Grad-CAM) (Selvaraju et al., 2017) to gain more insight into the learning of the network, which uses the gradients of each target concept that flow into the final convolutional layer to produce a coarse localization map that highlights the important regions in the image for predicting the concept.

The ERF of the central output unit is defined as the region containing any input pixel that has a non-negligible influence on that unit. In this study, we use backpropagation of gradients from the output of the target layer (usually the norm layer in the last convolution block) to the input image to estimate the size, shape, and offset of the resulting effective receptive field. (see Tables 2 and 3)

5. Results and discussions

5.1. Experimental results

Table 4 shows the crack segmentation results of the C4S dataset. Fig. 5 shows the visualization of the segmentation results, CAM and ERF on datasets with different training architectures. Based on this performance, important observations are offered.

5.2. Discussions

5.2.1. Impact of network depth on ERF

By comparing FCN models with different backbone depths, we found that a deeper backbone leads to better performance. FCN-8s with the ResNet-152 backbone performed better than FCN-8s with the ResNet-50 and ResNet-101 backbones. However, this improvement is limited. According to the visualization results, we can see that the deeper depth (e.g., from ResNet101 to ResNet-152) helps little increasing the ERFs, which means that the effective depth of the model has not increased significantly.

Table 2
Activation function.

Type	Formula	Advantages
Sigmoid	$f(x) = \frac{1}{(1+e^{-x})}$	Useful in binary classification tasks where the output needs to be a probability between 0 and 1.
Swish	$f(x) = x \cdot \text{sigmoid}(x)$	Swish allows data normalization and leads to quicker convergence .
Tanh	$f(x) = \frac{(e^x)-e^{-x})}{(e^x)+e^{-x})}$	Tanh maps any input value to a value between -1 and 1 with symmetric properties.
ReLU	$f(x) = \max(0, x)$	Sparsity, ease of calculation, and faster convergence.
Leaky ReLU	$f(x) = \max(ax, x)$	ReLU's variant that aims to prevent the "dying ReLU" problem.
ELU	$f(a, x) = \begin{cases} a(e^x - 1), & x < 0 \\ x & x \geqslant 0 \end{cases}$	Faster convergence and better accuracy.
GELU	$f(x) = x \cdot P(X < x)$	Widely used in Transformer, GeLU introduces a random regularity.

Table 3
Datased for training and evaluation.

Dataset	Sample Size	Resolution	Details
Crack500	500	2000 × 1500	captured by the cell phone with a pixel-level annotated binary map.
Crack260	260	800 × 600	captured by an area-array camera under visible light illumination
CrackSL315	331	512 × 512	captured by a line-array camera under laser illumination
CrackWH100	100	512 × 512	captured by a line-array camera under visible-light illumination
Stone331	500	1024 × 1024	captured by an area-array camera under visible-light illumination
C4S	2000	512 × 512	randomly sampled and cropped from the five crack datasets

Table 4
Segmentation performance of different methods.

Method	Backbone	Validation Set		Test Set	
		F1(%)	mIoU(%)	F1(%)	mIoU(%)
FCN-8s(1)	ResNet-50	66.41	58.36	62.35	55.34
FCN-8s(2)	ResNet-101	68.55	60.42	64.48	56.77
FCN-8s(3)	ResNet-152	69.69	60.90	63.69	54.23
U-Net	VGG-16	74.62	68.43	70.34	60.41
PSPNet	ResNet-50	76.38	69.72	71.45	62.89
UperNet	ResNet-50	78.56	71.81	72.30	67.32
Deeplabv3+	ResNet-50	77.41	71.56	71.44	66.43
HRNetv2	HRNetv2-18	82.19	74.14	77.65	68.72
HRNet-OCR	HRNetv2-18	**83.23**	74.98	**78.60**	**71.09**
RepLKNet	ReplkNet-31	82.64	**76.00**	77.36	70.31
Swin	Vision Transformer	71.49	63.28	65.34	56.21
Non-local	ResNet-50	75.30	67.77	69.22	61.05
Segformer	MixVisionTransformer	80.01	73.50	73.64	68.41
Segmenter	VisionTransformer	81.76	75.18	75.24	71.09

5.2.2. Larger kernel matters in segmentation

In contrast to 3 × 3 small kernels, CNNs with large kernels possess much larger ERFs and higher shape bias than texture bias. According to the result shown, RepLKNet has the largest effective receptive field among all methods. From the class activation map, it can be observed that RepLKNet extracts the most contextual information from the image, which helps to achieve SOTA performance.

5.2.3. Transformer-based models: Limitation of dataset

As shown in Fig. 5, all Transformer-based models possess enough ERFs to cover almost the entire image, which helps to learn more details of the crack. However, their performance on the C4S dataset is inferior to HRNet-OCR. The main reason is not that Transformer's structure is inferior to CNNs, but that Transformer's excellent performance relies more on the training sample size.

5.2.4. Significance of feature fusion and attention mechanism

HRNet-OCR performs best due to the combination of the HRNet backbone and the Object-Contextual Representations module, highlighting the importance of feature fusion and attention mechanism. HRNet achieves effective use of contextual information through excellent multi-branch architecture and feature fusion of different resolution branches. The OCR module learns the relationship between pixels and object region features based on the coarse segmentation results to improve the description of pixel features.

6. Conclusions and suggestions

DL approaches have become mainstream technologies for the development of automated pavement crack detection algorithms. To the best of our knowledge, this is the first review paper in the literature that focuses on the architecture construction of crack segmentation models. Our contribution is to highlight the critical factors that contribute to crack segmentation and to provide some suggestions to researchers on how to construct an intelligent and robust model.

Based on the critical analysis mentioned above, the following conclusions are summarized:

- In terms of both accuracy and speed, DL crack detection techniques outperform other types of methods. Semantic segmentation of cracks represents a fast, reliable, and repeatable approach to detecting cracks and has become the mainstream of automated crack recognition.

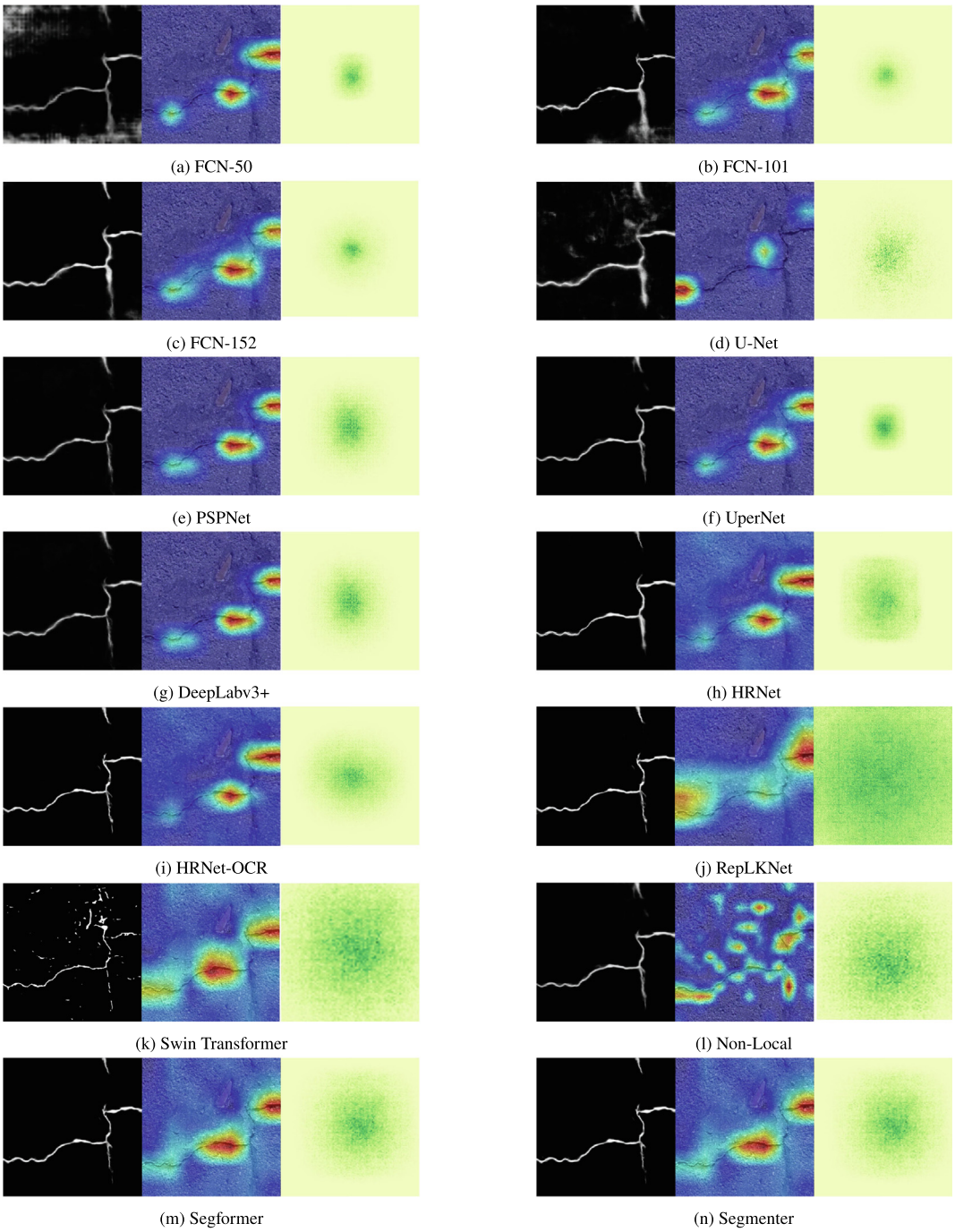


Fig. 5. Visualization results of the included models. Each sub-figure from left to right is a crack segmentation result, the visualization of the CAM, and the ERFs. According to the visualization, the larger the ERFs, the more effective the representation learned by the model as reflected by the CAM, and the more accurate the segmentation results.

- For crack segmentation tasks where training samples are limited and pixel-level labeling is costly, CNNs models have shown superior performance to the transformer-based model. There is enormous potential for combining CNNs with attention modules.
- A crack segmentation model should properly handle the conflict between classification and localization. According to the crack segmentation result and the visualization of ERFs and CAM, the key to improving the model performance is to expand the effective receptive field by feature fusion, attention mechanisms, and large convolution kernel.

- In addition to model structure design, image resolution, and training sample size affect model performance. The choice of appropriate evaluation metrics also serves to reflect the training results.

Cracks are often thin, irregular, and elongated, requiring models to possess sufficient ERFs and contextual information to segment them correctly. Crack segmentation algorithms need to be further developed in this direction to withstand the various disturbances faced when detecting cracks in complex environments. The application of appropriate activation functions and evaluation metrics also has a significant impact on model performance. Moreover, the current challenges faced by crack segmentation primarily stem from the poor generalization of models trained on small-scale datasets. Acquiring a large-scale dataset with high-quality annotations is a time-consuming and expensive task, therefore, automated pixel-level labeling methods for crack images deserve further exploration. With limited data, developing semi-supervised learning or weakly-supervised learning crack segmentation models makes sense.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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